WSMA LAB PROJECT · CSE_3132

Partial-Input Baselines Reveal Construction Artifacts in Fake News Benchmarks

I built a fake news detector, got 98%, then proved the 98% is fake.

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THE BENCHMARK

ISOT's two classes come from two newsrooms

21,417

real articles

All from Reuters, a wire agency with a strict house style

23,481

fake articles

From partisan websites flagged as unreliable

So a model can get the label right by recognising **who wrote the article**, without judging **whether it is true**.

Like a cancer detector that learns which hospital took the scan, then fails at a new hospital.

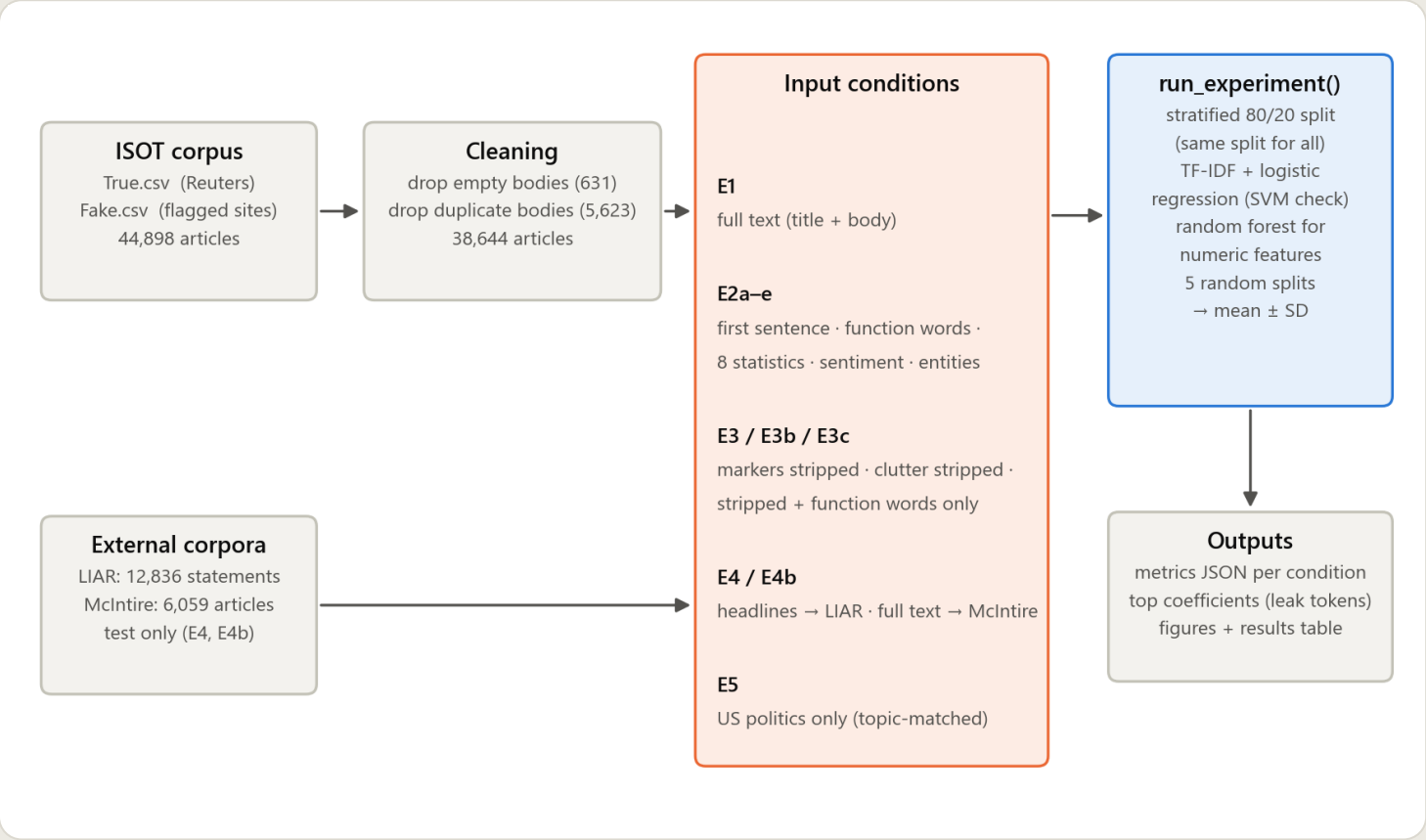
THE QUESTION

Is the detector recognising lies, or the newsroom?

Partial-input baselines (Gururangan et al., 2018): train models that are *not allowed* to see the article's content. If they still score far above chance, the dataset is leaking the answer.

METHOD

One harness, so only the input changes



- Cleaned ISOT: 38,644 articles (631 empty and 5,623 duplicate bodies removed)
- Same stratified 80/20 split, seed 42, for every condition
- TF-IDF + logistic regression; random forest for numeric features
- Five random splits per condition, reported as mean ± SD
- Chance measured, not assumed: 54.8%

EVIDENCE 1 · TAKE THE CONTENT AWAY

Without the content, models still score 86–99%

98.4%

Full text: the detector itself (E1)

99.1%

First sentence only: higher than the full article (E2a)

96.8%

Function words only: “the of a and to , the -” (E2b)

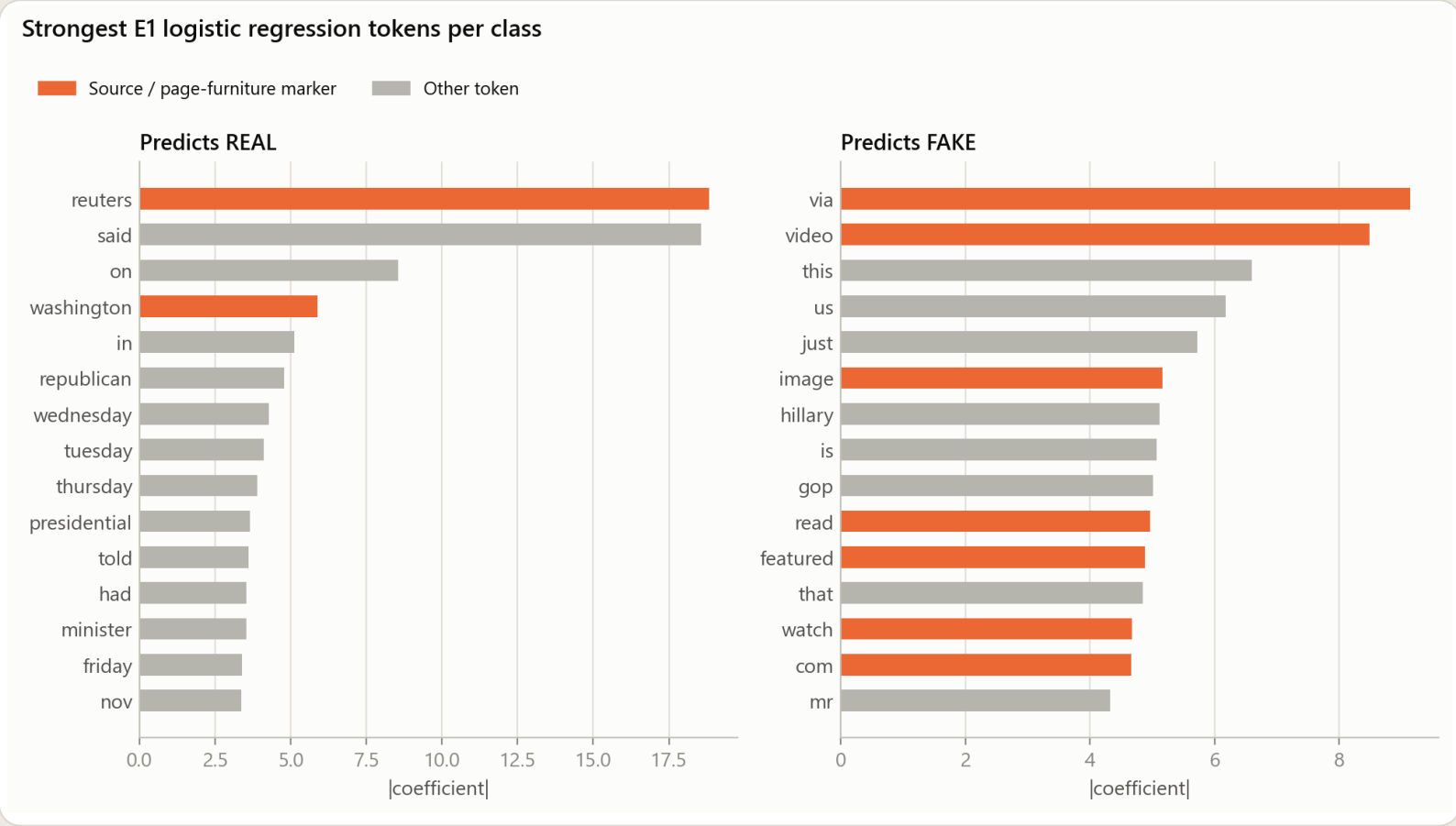
86.2%

Eight numbers, no words: length, “?”, “!”, capitals (E2c)

Chance (always guessing the larger class) is **54.8%**. Every probe is far above it.

EVIDENCE 2 · WHAT THE DETECTOR LEARNED

Its strongest clues name the publisher



Orange bars are source and page-clutter markers.

Real: “reuters”, “washington”, weekdays, “said”.

Fake: “via”, “image”, “featured”, “video”, “read”.

Who wrote it, not whether it is true.

EVIDENCE 3 · REMOVE THE GIVEAWAYS

Strip every marker from both classes: 97.3%

Accuracy as markers are removed

Condition	Accuracy
Full text (E1)	98.4%
Dateline, lead and “Reuters” removed (E3)	97.7%
+ credits, links, tags, quote marks (E3b)	97.3%
Stripped, function words only (E3c)	90.7%

What still separates the classes

REAL: WIRE REPORTING

said · told · reporters · spokesman · would

FAKE: BLOG VOICE

this · just · you · here · ! · ?

The source is not in a few words. It runs through the whole writing style.

Not the topic, and not the classifier

Topic: US politics only, both classes

90.3%

Every marker stripped, function words only (E5c), against **60.9%** chance on this subset.

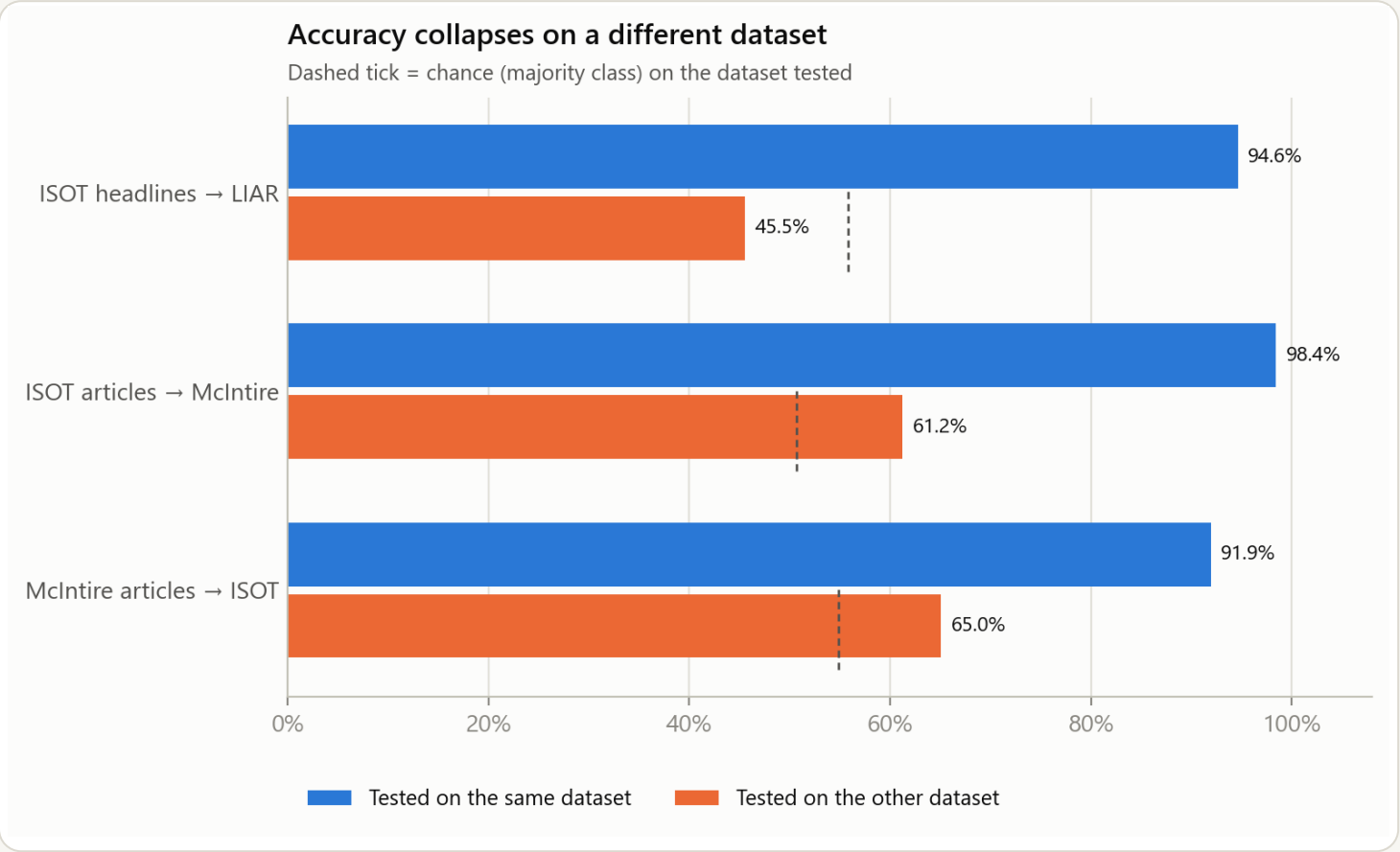
Real world-news articles removed, so topic cannot separate the classes.

Classifier: linear SVM instead

Input	Log. reg.	SVM
Full text	98.4%	99.3%
First sentence	99.1%	99.7%
Function words	96.8%	97.5%
All stripped	97.3%	98.1%

A stronger model exploits the shortcut slightly **more**.

On other datasets, the detector collapses



45.5%

on LIAR fact-checked statements, **below** its 55.8% chance

61.2%

on McIntire full articles (50.7% chance); 68% of genuine articles called fake

McIntire's own model scores **91.9%**, so the task there is learnable.

THE VERDICT

The 98% measures the newsroom, not the truth.

1

Report partial-input baselines next to every accuracy figure

2

Test on an independently built corpus, or on unseen publishers

3

Measure chance on the actual test set instead of assuming 50%

LIMITATIONS AND FUTURE WORK

What this does not show, and what comes next

Limitations

- Probes run in depth on ISOT only
- Classical models only, no transformers
- Hand-written stripping rules, extended once and disclosed

Future work

- Test sets from publishers never seen in training
- Benchmarks with true and false news from the same outlets
- The same probes on transformer models and other corpora

Code, data and results, reproducible in one command (about 5 minutes):
github.com/FirezTheGreat/fake-news-partial-input-baselines

Questions?